

# Tendon heals to scar by default.

## Nerves are what hold it back.

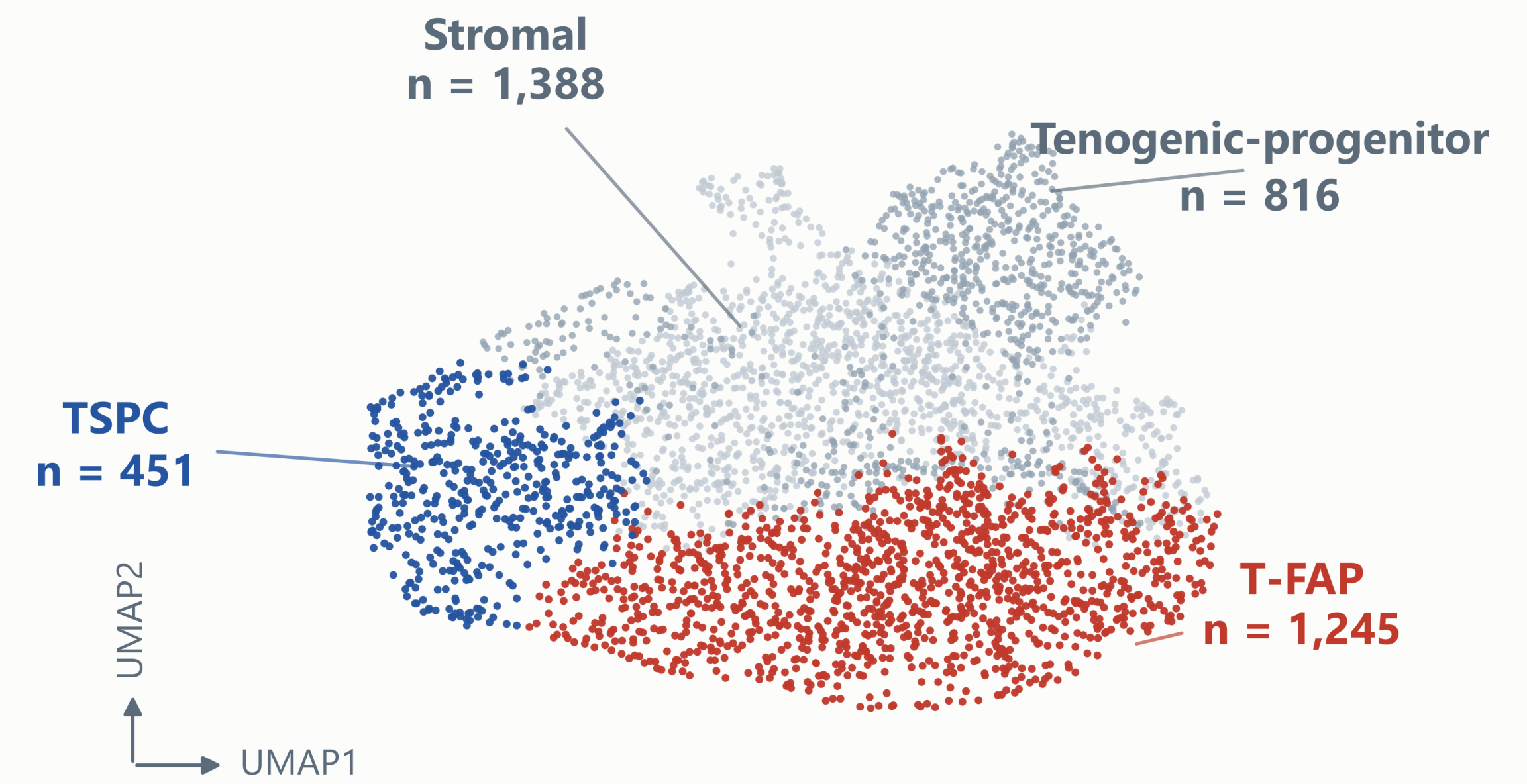
Mapping the transcription-factor program behind the regenerative-vs-fibrotic fate decision in tendon progenitors — and compressing it into a validated 23-gene Healing Index.

Fangyi Li<sup>1</sup> · Wilder Scott<sup>2</sup> · Yan Yan<sup>3</sup>

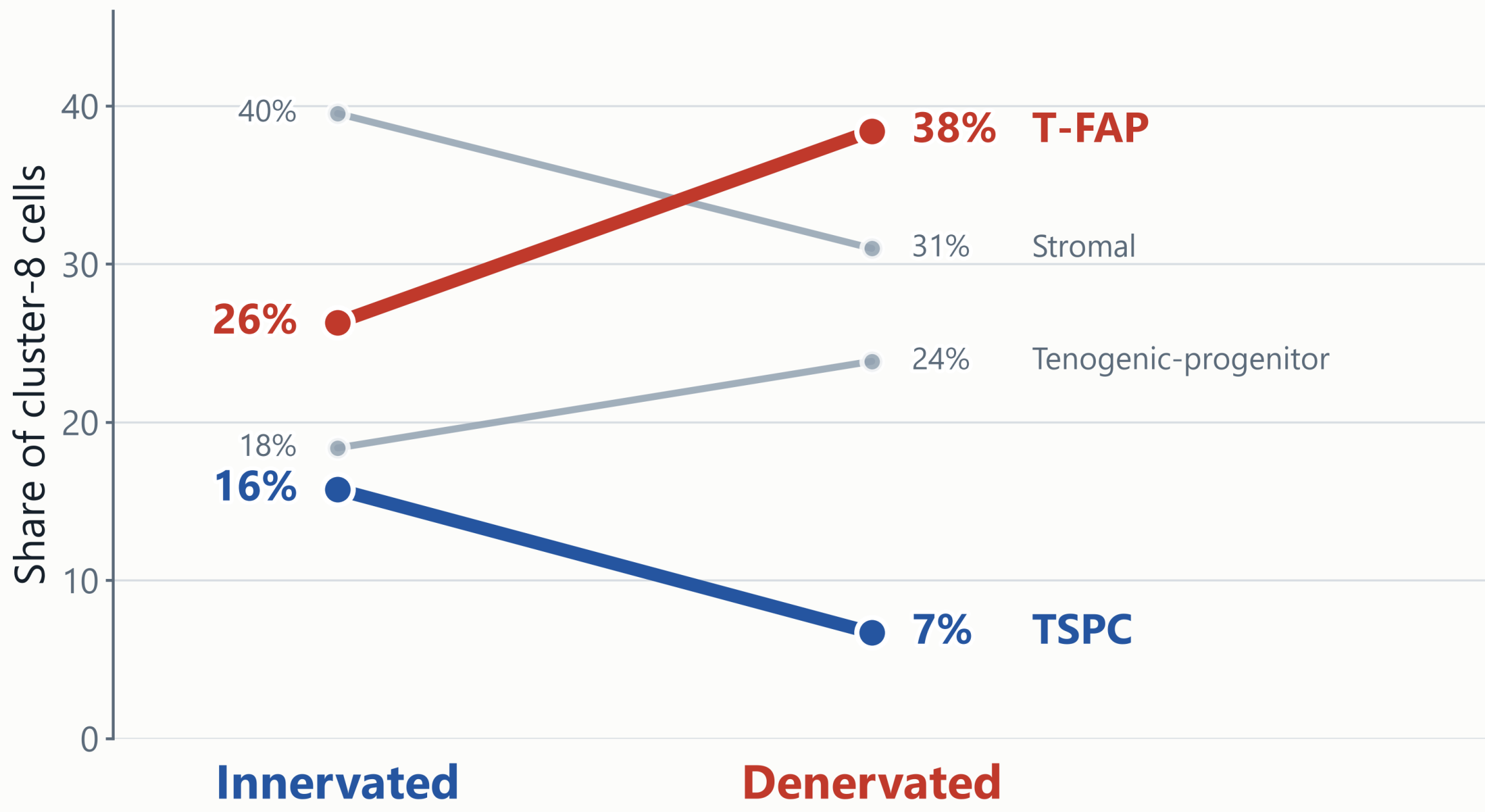
<sup>1</sup>Department of Integrative Biology, College of Biological Science, University of Guelph · <sup>2</sup>Sunnybrook Research Institute · <sup>3</sup>School of Computer Science, College of Computational, Mathematical and Physical Sciences, University of Guelph

### 1 Two cells, one decision

Tendon stem/progenitor cells (TSPC) rebuild aligned tendon. Tendon fibro-adipogenic progenitors (T-FAP) lay down scar instead. Both live in the same PDGFRα<sup>+</sup> pool.



3,900 PDGFRα<sup>+</sup> cells from injured mouse tendon.  
Sub-clustering resolves four populations (Cherief et al., 2023; day 14 post-injury).

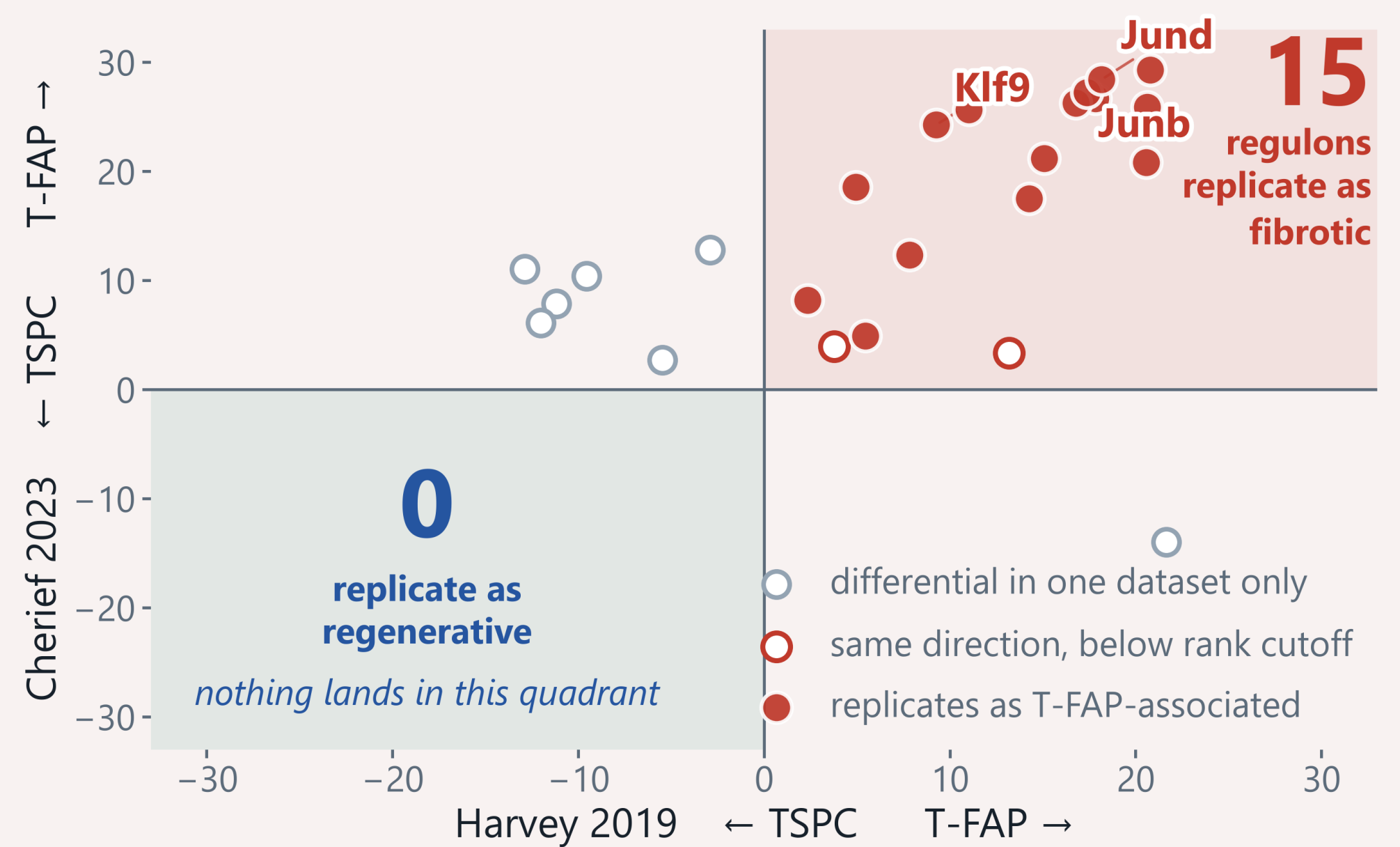


Blocking sensory-nerve signalling shifts the pool.  
TSPC share falls by more than half; T-FAP becomes the largest population. Nerve signalling is blocked chemical-genetically (TrkA-F592A mice + 1NMPP1), not surgically — “denervated” is shorthand throughout.

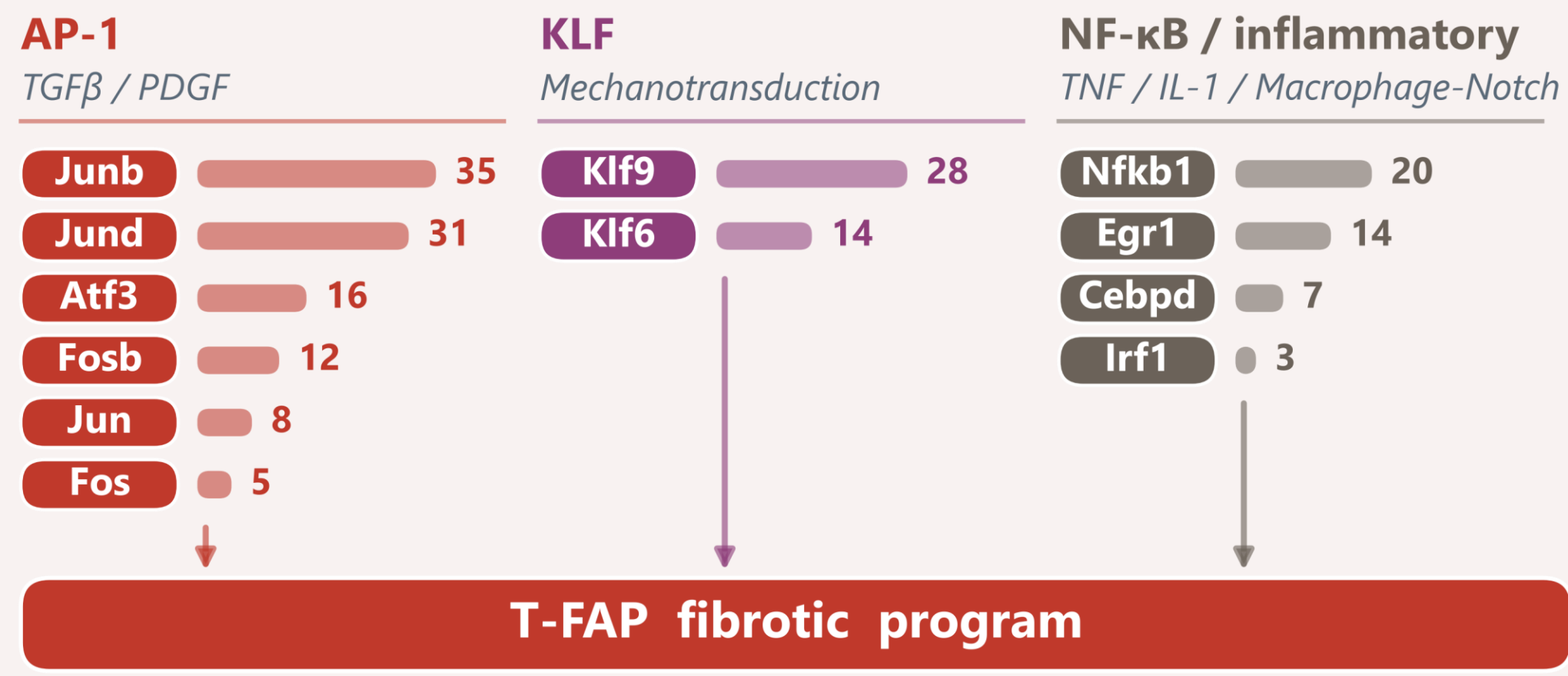
Innervation controls the outcome. The gene program that carries it out had never been mapped.

### 2 We looked for a repair switch. There wasn't one.

Which transcription-factor regulons — a TF plus the genes it directly regulates — separate the two fates? Networks were inferred independently in each dataset with pySCENIC (Van de Sande et al., 2020), then compared.



Fifteen regulons replicate as fibrotic. None replicate as regenerative.  
A 200-iteration power-matched audit reproduced the empty quadrant, so it is not a sample-size artefact.



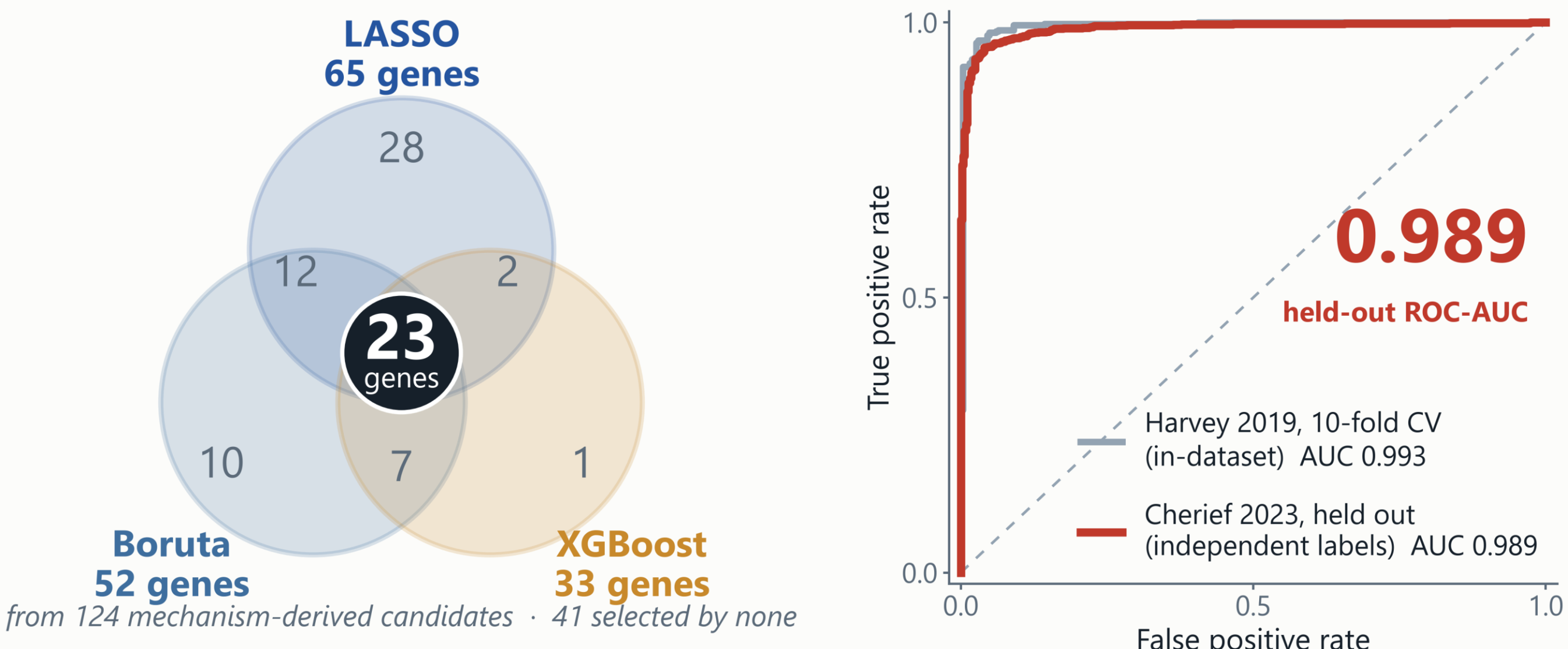
The surviving program is an AP-1 / KLF / NF-κB network.  
Junb, Jund and Klf9 contribute the most fate-driver genes (CellRank; Weiler et al., 2024); 14 of the 23 final panel genes are their targets. 12 of the 15 regulons are shown — three contribute none.

Fibrosis is not what happens when repair fails.  
Fibrosis is the default — and innervation is what suppresses it.

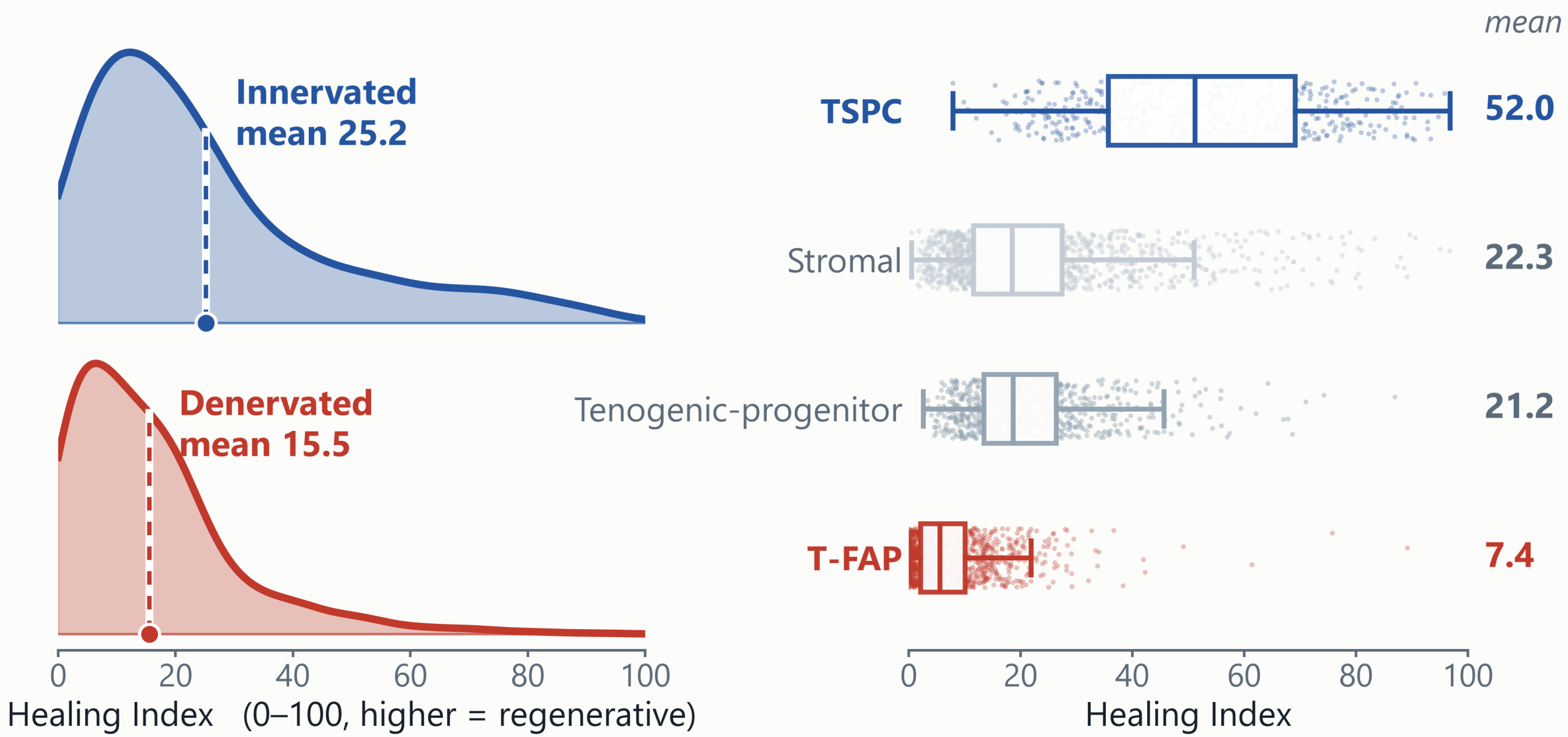
The mirror image of our starting hypothesis, which predicted an innervation-activated regenerative program.

### 3 So we measured the default

The fibrotic program is the one with reproducible signal, so it is also the one worth turning into a score.



Three methods agree on 23 genes; the panel then generalises.  
LASSO, Boruta and XGBoost ran independently on 124 mechanism-derived candidates. Trained on Harvey et al. (2019) and applied unchanged to Cherief et al. (2023), whose labels were assigned independently from marker expression and never seen during selection or training.



The score tracks innervation status.  
It also places two cell identities it never saw in training (Stromal, Tenogenic-progenitor) between the two fates.

#### We stress-tested our own headlines.

Two of the three did not survive — and saying so is what makes the third worth trusting.

"Efna1—Epha3 is the strongest differential signal"

LIANA permutation test, n = 100 (Dimitrov et al., 2022)

RETRACTED · p = 1.0

"T-FAP regulons feed back on Pdgfra"

Hypergeometric enrichment, BH-corrected

RETRACTED · nothing at p < 0.05

"Zero TSPC regulons replicate across datasets"

200× power-matched stability selection

CONFIRMED · still zero at matched n

#### WHAT TO TAKE AWAY

Healing a tendon may be less about switching regeneration on than about keeping the fibrotic default switched off. The Healing Index turns that idea into a number — ready to run on new tendon data the day it exists.

#### LIMITATIONS · WHAT'S NEXT

- Finding no reproducible regenerative regulon is not proof that none exists — both datasets are small, and a weaker TSPC program could be missed.
- Neither dataset has biological replicates — every validation so far is cross-dataset, not cross-animal.
- TSPC recall is 77–78% against 98–99% for T-FAP: the index flags fibrotic drift more reliably than it confirms full repair.
- Next: score newly generated lab tendon scRNA-seq — the first true biological-replicate test.

#### Data & code availability

All data are publicly available (GEO GSE244921; SRA PRJNA506218). No unpublished or confidential data were used. Computation was performed on the Digital Research Alliance of Canada (NIBI). Code and analysis notebooks: [github.com/lifangy6/sunnybrook-tendon-fate-mapping](https://github.com/lifangy6/sunnybrook-tendon-fate-mapping)

#### References

- Cherief, M., et al. (2023). TrkA-mediated sensory innervation of injured mouse tendon supports tendon sheath progenitor cell expansion and tendon repair. *Science Translational Medicine*, 15(727), eade4619.
- Van de Sande, B., et al. (2020). A scalable SCENIC workflow for single-cell gene regulatory network analysis. *Nature Protocols*, 15, 2247–2276.
- Dimitrov, D., et al. (2022). Comparison of methods and resources for cell-cell communication inference from single-cell RNA-Seq data. *Nature Communications*, 13, 3224.
- Weiler, P., Lange, M., Klein, M., Pe'er, D., & Theis, F. J. (2024). CellRank 2: Unified fate mapping in multiview single-cell data. *Nature Methods*, 21, 1196–1205.

#### Acknowledgements

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- Harvey, T., Flamenco, S., & Fan, C.-M. (2019). A Tppp3+Pdgfra<sup>+</sup> tendon stem cell population contributes to regeneration and reveals a shared role for PDGF signalling in regeneration and fibrosis. *Nature Cell Biology*, 21, 1490–1503.